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TRANSPORT CARRIER DOCKING DEVICE

Abstract

A transport carrier docking device may be capable of forming an air-tight seal around a transport carrier while a front portion of the transport carrier is inserted into a chamber of the transport carrier docking device. Semiconductor wafers in the transport carrier may be accessed by a transport tool while the air-tight seal exists around the transport carrier, which prevents and/or reduces the likelihood that contaminants in a semiconductor fabrication facility will reach the semiconductor wafers. The air-tight seal around the transport carrier may reduce defects of the semiconductor wafers that might otherwise be caused by the contaminants, may increase manufacturing yield and quality in the semiconductor fabrication facility, and/or may permit the continued reduction in device and/or feature sizes of integrated circuits and/or semiconductor devices that are to be formed on semiconductor wafers.

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Background/Summary

RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/657,423, filed Mar. 31, 2022, which is a continuation of U.S. patent application Ser. No. 17/248,072, filed Jan. 7, 2021 (now U.S. Pat. No. 11,302,553), the contents of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] A semiconductor wafer may be processed in various semiconductor processing tools in a semiconductor fabrication facility to produce various integrated circuits and/or semiconductor devices. A semiconductor wafer may be transported throughout the semiconductor fabrication facility and/or between the semiconductor processing tools in the semiconductor fabrication facility.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIG. 1 is a diagram of an example semiconductor processing environment described herein.

[0005] FIGS. 2A-2F are diagrams of example implementations of a sealing component for forming an air-tight seal around a transport carrier described herein.

[0006] FIG. 3 is a diagram of an example chamber door described herein.

[0007] FIGS. 4A-4N are diagrams of one or more example implementations described herein.

[0008] FIG. 5 is a diagram of example components of one or more devices of FIG. 2.

[0009] FIGS. 6 and 7 are flowcharts of example processes relating to accessing a transport carrier.

DETAILED DESCRIPTION

[0010] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments

and/or configurations discussed.

[0011] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0012] A plurality of semiconductor wafers and/or other types of substrates may be transported throughout a semiconductor fabrication facility in a transport carrier. A transport carrier may include a wafer cassette, a front-opening unified pod (FOUP), a pod, a container, or a similar type of device. To transfer a semiconductor wafer from a transport carrier to a semiconductor processing tool, the transport carrier may be placed in and/or on a load port associated with the semiconductor processing tool. A transport tool included in an interface tool (e.g., an equipment front end module (EFEM) or similar type of interface tool) that is situated between the semiconductor processing tool and the load port may remove the semiconductor wafer from the transport carrier. The transport tool may transfer the semiconductor wafer from the transport carrier to the semiconductor processing tool through a chamber of the interface tool. The transport tool may perform the above-described process in reverse to transfer the semiconductor wafer from the semiconductor processing tool to the transport carrier after processing.

[0013] Semiconductor wafers may be exposed to contaminants (e.g., volatile organic compounds (VOCs), dust, debris, and other types of contaminants) in the semiconductor fabrication facility during transfer of the semiconductor wafers between a load port and an associated semiconductor processing tool. These contaminants may cause semiconductor device failures, may cause defects to occur in the integrated circuits and/or semiconductor devices formed in the semiconductor fabrication facility, and may reduce manufacturing yield and quality, among other examples. Moreover, the impact of the contaminants in the semiconductor fabrication facility may continue to become more significant due to decreased tolerance to the contaminants as device and/or feature sizes of the integrated circuits and/or semiconductor devices, that are to be formed on semiconductor wafers in the semiconductor fabrication facility, continue to shrink.

[0014] Some implementations described herein provide a transport carrier docking device that can be positioned between a load port and an interface tool to reduce and/or minimize cross contamination of semiconductor wafers that are transferred between the load port and an associated semiconductor processing tool. The transport carrier docking device may be capable of forming an air-tight seal around a transport carrier while a front portion of the transport carrier is inserted into a chamber of the transport carrier docking device. In this way, semiconductor wafers in the transport carrier may be accessed by a transport tool while the air-tight seal around the transport carrier prevents and/or reduces the likelihood that contaminants in the semiconductor fabrication facility will reach the semiconductor wafers. Accordingly, the air-tight seal around the transport carrier may reduce defects of the semiconductor wafers that might otherwise be caused by the contaminants, may increase manufacturing yield and quality in the semiconductor fabrication facility, and/or may permit the continued reduction in device and/or feature sizes of integrated circuits and/or semiconductor devices that are to be formed on semiconductor wafers.

[0015] FIG. 1 is a diagram of an example semiconductor processing environment **100** described herein. The example semiconductor processing environment **100** may include, or may be included in, a semiconductor fabrication facility, a semiconductor foundry, a semiconductor processing facility, a semiconductor clean room, and/or another environment in which semiconductor wafers and/or devices are processed. As shown in FIG. 1, the example semiconductor processing environment **100** may include a semiconductor processing tool **102**, a load port **104**, an interface tool **106**, and a docking device **108**, among other tools and/or devices.

[0016] The semiconductor processing tool **102** may include one or more tools configured to perform one or more semiconductor processing operations on one or more semiconductor wafers and/or devices. For example, the semiconductor processing tool **102** may include a deposition tool (e.g., a semiconductor processing tool configured to deposit one or more layers onto a semiconductor wafer), a plating tool (e.g., an electroplating tool configured to deposit one or more metal layers onto a semiconductor wafer), an exposure tool (e.g., an extreme ultraviolet (EUV) tool, an electron beam (e-beam) tool), an etch tool (e.g., a wet etch tool, a dry etch tool), or another type of semiconductor processing tool.

[0017] The load port **104** may include a shuttle platform **110** configured to receive and support a transport carrier **112**. The transport carrier **112** may include a wafer cassette, a FOUP, a pod, a container, or a similar type of device configured to hold and/or store a plurality of semiconductor wafers. The load port **104** and the shuttle platform **110** may receive the transport carrier **112** from a transport robot, a transport cart, an overhead hoist transport (OHT), or another device configured to move transport carriers to and from various locations in the example semiconductor processing environment **100**. The transport carrier **112** may include a body **114** configured to rest on the shuttle platform **110**. The load port **104** may further include an exhaust **116** that is configured to vent the load port **104**.

[0018] The interface tool **106** may include an EFEM or another tool that includes a chamber **118**. Air may be provided to the chamber **118** through a filter **120** (e.g., a high-efficiency particulate air (HEPA) filter or another type of air filter) configured to filter the incoming air of particles and other contaminants. The chamber **118** may further be vented through an exhaust **122**.

[0019] The interface tool **106** may further include a wafer transport tool **124** in the chamber **118**. The wafer transport tool **124** may include a robotic arm or another type of tool that is configured to transport semiconductor wafers between the transport carrier **112** and the semiconductor processing tool **102**. In some implementations, the wafer transport tool **124** transfers semiconductor wafers between the transport carrier **112** and a staging area of the semiconductor processing tool **102**. In some implementations, the wafer transport tool **124** transfers semiconductor wafers between the transport carrier **112** and a processing chamber of the semiconductor processing tool **102**.

[0020] The docking device **108** is a device that is configured to permit the transfer of semiconductor wafers between the transport carrier **112** and the semiconductor processing tool **102** in a manner that reduces the likelihood of exposure of semiconductor wafers to contaminants that may be present in the example semiconductor processing environment **100**. The docking device **108** may include an opening **126** at a first side (e.g., a first sidewall) of the docking device **108** and another opening **128** at a second side (e.g., a second sidewall) of the docking device **108**. The wafer transport tool **124** may access the transport carrier **112** (e.g., the body **114** of the transport carrier **112**) through the opening **126** and the opening **128**. The wafer transport tool **124** may transport semiconductor wafers to and from the semiconductor processing tool **102** through an opening **130** between the interface tool **106** and the semiconductor processing tool **102**. In some implementations, the opening **126** and the opening **128** are on opposing sides (e.g., opposing sidewalls) of the docking device **108**. In some implementations, the opening **126** and the opening **128** are on adjacent sides (e.g., adjacent sidewalls) of the docking device **108**.

[0021] The docking device **108** may include a chamber **132**. The opening **126** may provide access to the chamber **132** for insertion of the transport carrier **112**, and the opening **128** may provide access to the chamber **132** for the wafer transport tool **124**. The chamber **132** may include a door frame **134**, a chamber door **136**, and a sealing component **138** to reduce, minimize, and/or eliminate the likelihood of exposure to contaminants that may be present in the example semiconductor processing environment **100** when a semiconductor wafer is transferred between the transport carrier **112** and the semiconductor processing tool **102**.

[0022] The door frame **134** may include a rigid or semi-rigid structure that extends from the top of the chamber **132** to the bottom of the chamber **132**, and that extends from one side of the chamber

132 to another (opposing) side of the chamber **132**. An opening may be provided through the door frame **134**, which may be sealed by the chamber door **136**. In particular, the chamber door **136** may be pressed against the door frame **134** to form an air-tight seal between the door frame **134** and the chamber door **136**. The door frame **134** and the chamber door **136** may form the air-tight seal when the chamber door **136** is in a closed position, as shown in FIG. **1**. The air-tight seal may reduce, minimize, and/or eliminate the likelihood that contaminants in the example semiconductor processing environment **100** might travel through the opening **126** and through the opening in the door frame **134** when the transport carrier **112** is not inserted into the opening **126** and/or when no air-tight seal is formed around the transport carrier **112**.

[0023] In some implementations, the door frame **134** and/or the chamber door **136** may include a gasket, a strip, or another component to form the air-tight seal. The gasket of the door frame **134**, if included, may be formed around the opening in the door frame **134**. The gasket of the chamber door **136**, if included, may be formed in a shape that substantially fits around the opening in the door frame **134**. The gasket of the door frame **134** and/or the gasket of the chamber door **136** may be formed of a soft material and/or of a deformable material to permit the air-tight seal to be formed. For example, the gasket of the door frame **134** and/or the gasket of the chamber door **136** may be formed of a plastic material, a rubber material, a silicone material, or a similar material. In some implementations, the material of the gasket of the door frame **134** and/or the gasket of the chamber door **136** includes a gas impermeable material.

[0024] The sealing component **138** may be pressed against the door frame **134** on an opposite side of the door frame **134** as the chamber door **136**. One or more gaskets may be located between the sealing component **138** and door frame **134** to reduce, minimize, and/or prevent air (and contaminants carried by the air) from passing between the sealing component **138** and the door frame **134**. Moreover, the sealing component **138** may be pressed against the side (e.g., the sidewall) of the chamber **132** on which the opening **126** is located such that an air-tight seal is formed between the side and the sealing component **138**.

[0025] The sealing component **138** may include one or more portions (e.g., portion **138a**, portion **138b**, and/or one or more other portions) that are configured to form an air-tight seal around the transport carrier **112** when the transport carrier **112** is at least partially inserted into the opening **126**. For example, the shuttle platform **110** may slide or otherwise move the transport carrier **112** toward and at least partially into the opening **126**. With the transport carrier **112** at least partially inserted through the opening **126** and into the chamber **132**, the sealing component **138** may contract around the transport carrier **112** (e.g., around the body **114** of the transport carrier **112**) to form the air-tight seal between the sealing component **138** and the transport carrier **112**.

[0026] In some implementations, the sealing component **138** (or the portions thereof) are formed of a relatively soft material (e.g., softer than the material of the body **114** of the transport carrier **112**) to permit the air-tight seal to be formed around the transport carrier **112**. For example, the sealing component **138** (or the portions thereof) may be formed of a plastic material, a rubber material, a silicone material, a gas impermeable, or a similar material. In some implementations, the sealing component **138** (or the portions thereof) includes a gasket, a strip, or a similar component formed of a relatively soft material that is pressed against the body **114** of the transport carrier **112** to form the air-tight seal.

[0027] When the air-tight seal is formed between the sealing component **138** and the transport carrier **112**, the chamber door **136** may remove a carrier door **140** from the body **114** of the transport carrier **112**. After removing the carrier door **140**, the chamber door **136** may transition from the closed position shown in FIG. **1** to an open position, which releases the air-tight seal between door frame **134** and the chamber door **136**. Here, the chamber door **136** (with the removed carrier door **140**) may move backwards away from the door frame **134** toward the opening **128**, and may move downward into the chamber **132** (e.g., after moving backwards away from the door frame **134**). In this way, the opening in the door frame **134** is cleared such that the wafer transport

tool **124** is permitted to access the transport carrier **112** (e.g., the body **114** of the transport carrier **112**) through the opening in the door frame **134** and the opening **128**.

[0028] The wafer transport tool **124** may transfer semiconductor wafers between the transport carrier **112** and the semiconductor processing tool **102** after the air-tight seal is formed around the transport carrier **112** by the sealing component **138**, and after the chamber door **136** is opened. For example, the wafer transport tool **124** may obtain and/or retrieve a semiconductor wafer from the transport carrier **112** through the opening in the door frame **134** and through the opening **128**, and may provide the wafer to the semiconductor processing tool **102** through the opening **130**. As another example, the wafer transport tool **124** may obtain a semiconductor wafer from the semiconductor processing tool **102** through the opening **130**, and may provide the semiconductor wafer to the transport carrier **112** through the opening **128** and the opening in the door frame **134**.

[0029] In this way, with the carrier door **140** removed and the chamber door **136** in the open position, air-tight seals are formed between the sidewall on which the opening **126** is located and the sidewall on which the opening **128** is located. In particular, an air-tight seal may be formed between the transport carrier **112** and the sealing component **138**, an air-tight seal may be formed between the sealing component **138** and the wall of the chamber **132** on which the opening **126** is located, an air-tight seal may be formed between the sealing component **138** and the door frame **134**, and an air-tight seal may be formed between the door frame **134** and the top, bottom, and sides of the chamber **132**. This provides an air-tight seal through the chamber **132** between the opening **126** and the opening **128**. The air-tight seal through the chamber **132** between the opening **126** and the opening **128** permits the transport carrier **112** to be accessed without exposing the semiconductor wafers that are between the transport carrier **112** and the semiconductor processing tool **102**.

[0030] As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described with regard to FIG. **1**.

[0031] FIGS. **2A-2F** are diagrams of example implementations of the sealing component **138** for forming an air-tight seal around a transport carrier **112** described herein. FIG. **2A** illustrates perspective views of example implementations **210** and **220** of the sealing component **138** in which the sealing component **138** includes two portions: a portion **138a** and portion **138b**. Example implementation **210** illustrates an example of the sealing component **138** in an expanded configuration, where the portion **138a** and the portion **138b** are not contracted around the body **114** of the transport carrier **112**. In this configuration, the portion **138a** and the portion **138b** may be spaced away from the body **114** of the transport carrier **112** at a distance that permits the carrier door **140** and the body **114** of the transport carrier **112** to fit through the opening formed by the portion **138a** and the portion **138b**.

[0032] As further shown in the example implementation **220** in FIG. **2A**, the sealing component **138** may transition to a contracted configuration, in which the portion **138a** and the portion **138b** are contracted around the body **114** of the transport carrier **112** to form the air-tight seal around the transport carrier **112**.

[0033] FIG. **2B** illustrates an elevation view of an example implementation **230**, which shows a first position **240** of the portions **138a** and **138b**, and a second position **250** of the portions **138a** and **138b**. The first position **240** may correspond to the expanded configuration illustrated in the example implementation **210** of FIG. **2A**, and the second position **250** may correspond to the contracted configuration illustrated in the example implementation **220** of FIG. **2A**. As shown in FIG. **2B**, in some implementations, the portion **138a** and the portion **138b** may each transition between the first position **240** and the second position **250** in a diagonal path of travel. For example, the portion **138a** may move downward and inward in a diagonal path of travel from the first position **240** to the second position **250**, and may move upward and outward in a diagonal path of travel from the second position **250** to the first position **240**. As another example, the portion **138b** may move upward and inward in a diagonal path of travel from the first position **240** to the

second position **250**, and may move downward and outward in a diagonal path of travel from the second position **250** to the first position **240**.

[0034] As further shown in FIG. 2B, in the example implementation **230**, the portion **138a** may cover substantially the entire top of the transport carrier **112** and substantially an entire first side of the transport carrier **112**. The portion **138b** may cover substantially the entire bottom of the transport carrier **112** and substantially an entire second side of the transport carrier **112** opposing the first side.

[0035] FIG. 2C illustrates an elevation view of an example implementation **260**. In the example implementation **260**, the sealing component **138** includes a plurality of portions: a portion **138a** and a second portion **138b**. The portion **138a** may cover substantially the entire top of the transport carrier **112**, a first portion of a first side of the transport carrier **112**, and a first portion of a second side of the transport carrier **112** opposing the first side. The portion **138b** may cover substantially the entire bottom of the transport carrier **112**, a second portion of the first side of the transport carrier **112**, and a second portion of the second side of the transport carrier **112**. The first portion of the first side covered by the portion **138a**, and the second portion of the first side covered by the portion **138b** may substantially encompass the entire first side of the transport carrier **112**. The first portion of the second side covered by the portion **138a**, and the second portion of the second side covered by the portion **138b** may substantially encompass the entire second side of the transport carrier **112**.

[0036] FIG. 2D illustrates an elevation view of an example implementation **270**. In the example implementation **270**, the sealing component **138** includes a plurality of portions: a portion **138a** and a second portion **138b**. The portion **138a** may cover substantially an entire first side of the transport carrier **112**, a first portion of the top of the transport carrier **112**, and a first portion of the bottom of the transport carrier **112**. The portion **138b** may cover substantially an entire second side of the transport carrier **112** opposing the first side, a second portion of the top of the transport carrier **112**, and a second portion of the bottom of the transport carrier **112**. The first portion of the top covered by the portion **138a**, and the second portion of the top covered by the portion **138b** may substantially encompass the entire top of the transport carrier **112**. The first portion of the bottom covered by the portion **138a**, and the second portion of the bottom covered by the portion **138b** may substantially encompass the entire bottom of the transport carrier **112**.

[0037] FIG. 2E illustrates an elevation view of an example implementation **280**. In the example implementation **280**, the sealing component **138** includes a plurality of portions: a portion **138a**, a portion **138b**, and a portion **138c**. The portion **138a** may cover a first portion of the top of the transport carrier **112** and substantially an entire first side of the transport carrier **112**. The portion **138b** may cover a second portion of the top side of the transport carrier **112** and substantially an entire second side of the transport carrier **112** opposing the first side. The portion **138c** may cover the entire bottom of the transport carrier **112**. Other example implementations of the sealing component **138** having three portions may be used with the docking device **108**.

[0038] FIG. 2F illustrates an elevation view of an example implementation **290**. In the example implementation **290**, the sealing component **138** includes a plurality of portions: a portion **138a**, a portion **138b**, a portion **138c**, and a portion **138d**. The portion **138a** may cover substantially the entire top of the transport carrier **112**. The portion **138b** may cover substantially the entire bottom of the transport carrier **112**. The portion **138c** may cover substantially an entire first side of the transport carrier **112**. The portion **138d** may cover substantially an entire second side of the transport carrier **112** opposing the first side. Other example implementations of the sealing component **138** having four portions may be used with the docking device **108**.

[0039] As indicated above, FIGS. 2A-2F are provided as examples. Other examples may differ from what is described with regard to FIGS. 2A-2F. In some implementations, the sealing component **138** may include five or more portions in various configurations different from those described herein.

[0040] FIG. 3 is a diagram of an example **300** of the chamber door **136** described herein. As shown in FIG. 3, the chamber door **136** may include an elongated body. Moreover, the chamber door **136** may include a mounting plate **302** attached, connected, and/or otherwise mounted to a support structure **304**. The support structure **304** may be attached, connected, and/or otherwise mounted to the elongated body of the chamber door **136**. The mounting plate **302** and the support structure **304** may be attached, connected, and/or otherwise mounted to a side of the elongated body of the chamber **132** that faces the door frame **134** in the chamber **132** of the docking device **108**.

[0041] The mounting plate **302** may be configured to remove the carrier door **140** from the body **114** of the transport carrier **112**, may be configured to hold and/or support the carrier door **140** of the transport carrier **112** when the chamber door **136** moves between the closed position and the open position, and/or may be configured to place the carrier door **140** onto the body **114** of the transport carrier **112**. The mounting plate **302** may include one or more vacuum holes **306** and one or more latch keys **308**. The vacuum holes **306** may be connected to tube(s), plumbing fixture(s), and/or one or more other components that are configured to pull air through the vacuum holes **306** to form a negative pressure on the face of the mounting plate **302** near the vacuum holes **306**. The negative pressure may form a suction or a vacuum seal between the mounting plate **302** and the carrier door **140**. The mounting plate **302** may hold and/or support the carrier door **140** based on the negative pressure generated through the vacuum holes **306**.

[0042] The latch key(s) **308** may be configured to unlatch the carrier door **140** from the body **114** of the transport carrier **112**. For example, the latch key(s) **308** may include a key, an elongated member, or another component that is configured to open a door latch and/or a door lock on the carrier door **14**. In some implementations, the mounting plate **302** includes a single latch key **308** (e.g., on a side of the mounting plate **302** or substantially near a center of the mounting plate **302**). In some implementations, the mounting plate **302** includes a plurality of latch keys **308** substantially near one or more edges of the mounting plate **302**.

[0043] The support structure **304** may include a jack (e.g., a scissor jack, a hydraulic jack, a pneumatic jack, or another type of jack), a screw mechanism, a rail system, or another type of structure configured to extend the mounting plate **302** away from the elongated body of the chamber door **136** and contract the mounting plate **302** toward the elongated body of the chamber door **136**. In this way, the support structure **304** may extend the mounting plate **302** toward the carrier door **140** to remove the carrier door **140** from the transport carrier **112**, may contract the mounting plate **302** after removing the carrier door **140** so that the chamber door **136** may be moved to the opened position, and may extend the mounting plate **302** toward transport carrier **112** to place the carrier door **140** onto the transport carrier **112**.

[0044] As indicated above, FIG. 3 is provided as an example. Other examples may differ from what is described with regard to FIG. 3.

[0045] FIGS. 4A-4N are diagrams of one or more example implementations **400** described herein. The example implementation(s) **400** may include one or more example implementations illustrating various operations of the docking device **108** and other tools included in the example semiconductor processing environment **100**. As shown in FIG. 4A, a transport carrier **112** may be placed on the shuttle platform **110** of the load port **104**. For example, a mobile robot, an OHT, or another transport tool may place the transport carrier **112** on the shuttle platform **110**.

[0046] As shown in FIG. 4B, the shuttle platform **110** may slide or otherwise move toward the opening **126** in the docking device **108** such that the transport carrier **112** is at least partially inserted into the chamber **132** of the docking device **108** through the opening **126**. The shuttle platform **110** may slide or otherwise move toward the opening **126** in the docking device **108** such that the carrier door **140** of the transport carrier **112** is fully inserted into the chamber **132** of the docking device **108** through the opening **126**, and such that the body **114** of the transport carrier **112** is at least partially inserted into the chamber **132** of the docking device **108** through the opening **126**.

[0047] As shown in the close-up view **402** in FIG. **4B**, the carrier door **140** clears the width or thickness of the sealing component **138** when the transport carrier **112** is at least partially inserted into the chamber **132**. As further shown in the close-up view **402** in FIG. **4B**, the portions **138a** and **138b** of the sealing component **138** may be in the first position **240** (e.g., the expanded configuration) when the transport carrier **112** is at least partially inserted into the chamber **132** of the docking device **108** through the opening **126** to provide clearance between the transport carrier **112** and the sealing component **138**.

[0048] As shown in FIG. **4C**, with the transport carrier **112** at least partially inserted into the chamber **132** through the opening **126**, the sealing component **138** may contract around the transport carrier **112** to form an air-tight seal around the transport carrier **112**. In particular, the air-tight seal may be formed between the sealing component **138** and the transport carrier **112**, and may be formed between the side of the chamber **132** (e.g., the sidewall) on which the opening **126** is located and the sealing component **138**. In this way, the opening **126** is sealed to reduce, minimize, and/or eliminate the flow of contaminants from the example semiconductor processing environment **100** through the opening **126** and into the chamber **132**. As shown in the close-up view **404** in FIG. **4C**, the portions **138a** and **138b** may contract around the transport carrier **112** by transitioning from the first position **240** (e.g., the expanded configuration) to the second position **250** (e.g., the contracted configuration).

[0049] As shown in FIG. **4D**, with the air-tight seal formed around the transport carrier **112** by the sealing component **138**, the chamber door **136** may remove the carrier door **140** from the transport carrier **112**. For example, the support structure **304** may extend the mounting plate **302** toward the carrier door **140** through the opening in the door frame **134**. The one or more latch keys **308** on the mounting plate **302** may unlatch the carrier door **140** from the body **114** of the transport carrier **112**. The vacuum holes **306** on the mounting plate **302** may form a vacuum seal between the carrier door **140** and the mounting plate **302**. With the vacuum seal formed, the support structure **304** may contract the mounting plate **302** (with the carrier door **140** supported and/or held thereon) toward the chamber door **136**.

[0050] As further shown in FIG. **4D**, the chamber door **136** may slide and/or otherwise move backward away from the door frame **134** and toward the opening **128**, which removes the air-tight seal between door frame **134** and the chamber door **136**. In particular, the chamber door **136** may slide and/or otherwise move backward away from the door frame **134** and toward the opening **128** after removing the carrier door **140** from the transport carrier **112** and contracting the mounting plate **302** toward the chamber door **136**. The chamber door **136** may slide and/or otherwise move backward away from the door frame **134** a sufficient distance to permit the carrier door **140** to clear the door frame **134** when the chamber door **136** slides and/or otherwise moves downward into the chamber **132**.

[0051] As shown in FIG. **4E**, the chamber door **136** may slide and/or otherwise move out of the way of the opening in the door frame **134** such that the wafer transport tool **124** is permitted to access the transport carrier **112** through the opening in the door frame **134**. In some implementations, the chamber door **136** slides and/or otherwise moves downward within the chamber **132**. In some implementations, the chamber door **136** slides and/or otherwise moves upward within the chamber **132**. In some implementations, the chamber door **136** slides and/or otherwise moves to a side within the chamber **132**. In some implementations, the chamber door **136** slides and/or otherwise moves in a combination of directions within the chamber **132**.

[0052] As shown in FIG. **4F**, the wafer transport tool **124** of the interface tool **106** may retrieve and/or obtain a wafer **406** from the transport carrier **112** through the opening in the door frame **134** and through the opening **128**. In this way, the wafer transport tool **124** retrieves and/or obtains the wafer **406** from the transport carrier **112** while the air-tight seal is formed around the transport carrier **112** by the sealing component **138**. As shown in FIG. **4G**, the wafer transport tool **124** may retract the wafer **406** through the opening in the door frame **134** and through the opening **128** into

the chamber **118** of the interface tool **106**. As shown in FIG. **4H**, the wafer transport tool **124** may provide the wafer **406** to the semiconductor processing tool **102** through the opening **130**. The wafer **406** may be processed by the semiconductor processing tool **102** through one or more semiconductor processing operations.

[0053] As shown in FIG. **4I**, the wafer transport tool **124** may retrieve the wafer **406** from the semiconductor processing tool **102** through the opening **130** (e.g., after the semiconductor processing tool **102** processes the wafer **406** through one or more semiconductor processing operations) and retracts the wafer **406** into the chamber **118** of the interface tool **106**. As shown in FIG. **4J**, the wafer transport tool **124** may extend into the opening **128** and the opening in the chamber door **136** to place and/or otherwise provide the wafer **406** to the transport carrier **112**.

[0054] As shown in FIG. **4K**, the chamber door **136** (with the carrier door **140** mounted to the mounting plate **302**) may move upward within the chamber **132**. As shown in FIG. **4L**, the chamber door **136** (with the carrier door **140** mounted to the mounting plate **302**) may move toward the door frame **134** and may form an air-tight seal between the door frame **134** and the chamber door **136**. The support structure **304** may extend toward the transport carrier **112** and may place the carrier door **140** onto the body **114** of the transport carrier **112**. The vacuum hole(s) **306** may release the vacuum seal between the carrier door **140** and the mounting plate **302**, and the latch key(s) **308** may latch the carrier door **140** onto the body **114** of the transport carrier **112**.

[0055] As shown in FIG. **4M**, the sealing components **138** may transition from the second position **250** (e.g., the contract configuration) to the first position **240** (e.g., the expanded configuration) to release the air-tight seal around the transport carrier **112**. As shown in FIG. **4N**, the shuttle platform **110** may slide and/or otherwise move the transport carrier **112** out of the chamber **132** and the opening **126**. In some implementations, a mobile robot, an OHT, or another transport tool may transport the transport carrier **112** to another location in the example semiconductor processing environment **100**.

[0056] As indicated above, FIGS. **4A-4N** are provided as examples. Other examples may differ from what is described with regard to FIGS. **4A-4N**.

[0057] FIG. **5** is a diagram of example components of a device **500**. In some implementations, one or more devices and/or tools of the example semiconductor processing environment **100** (e.g., the semiconductor processing tool **102**, the load port **104**, the interface tool **106**, and/or the docking device **108**) may include one or more devices **500** and/or one or more components of device **500**. As shown in FIG. **5**, device **500** may include a bus **510**, a processor **520**, a memory **530**, a storage component **540**, an input component **550**, an output component **560**, and a communication component **570**.

[0058] Bus **510** includes a component that enables wired and/or wireless communication among the components of device **500**. Processor **520** includes a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. Processor **520** is implemented in hardware, firmware, or a combination of hardware and software. In some implementations, processor **520** includes one or more processors capable of being programmed to perform a function. Memory **530** includes a random access memory, a read only memory, and/or another type of memory (e.g., a flash memory, a magnetic memory, and/or an optical memory).

[0059] Storage component **540** stores information and/or software related to the operation of device **500**. For example, storage component **540** may include a hard disk drive, a magnetic disk drive, an optical disk drive, a solid state disk drive, a compact disc, a digital versatile disc, and/or another type of non-transitory computer-readable medium. Input component **550** enables device **500** to receive input, such as user input and/or sensed inputs. For example, input component **550** may include a touch screen, a keyboard, a keypad, a mouse, a button, a microphone, a switch, a sensor, a global positioning system component, an accelerometer, a gyroscope, and/or an actuator. Output

component **560** enables device **500** to provide output, such as via a display, a speaker, and/or one or more light-emitting diodes. Communication component **570** enables device **500** to communicate with other devices, such as via a wired connection and/or a wireless connection. For example, communication component **570** may include a receiver, a transmitter, a transceiver, a modem, a network interface card, and/or an antenna.

[0060] Device **500** may perform one or more processes described herein. For example, a non-transitory computer-readable medium (e.g., memory **530** and/or storage component **540**) may store a set of instructions (e.g., one or more instructions, code, software code, and/or program code) for execution by processor **520**. Processor **520** may execute the set of instructions to perform one or more processes described herein. In some implementations, execution of the set of instructions, by one or more processors **520**, causes the one or more processors **520** and/or the device **500** to perform one or more processes described herein. In some implementations, hardwired circuitry may be used instead of or in combination with the instructions to perform one or more processes described herein. Thus, implementations described herein are not limited to any specific combination of hardware circuitry and software.

[0061] The number and arrangement of components shown in FIG. **5** are provided as an example. Device **500** may include additional components, fewer components, different components, or differently arranged components than those shown in FIG. **5**. Additionally, or alternatively, a set of components (e.g., one or more components) of device **500** may perform one or more functions described as being performed by another set of components of device **500**.

[0062] FIG. **6** is a flowchart of an example process **600** associated with accessing a transport carrier. In some implementations, one or more process blocks of FIG. **6** may be performed by a docking device (e.g., docking device **108**). In some implementations, one or more process blocks of FIG. **6** may be performed by another device or a group of devices separate from or including the docking device, such as a semiconductor processing tool **102**, a load port **104**, an interface tool **106**, a wafer transport tool **124**, and/or another device. Additionally, or alternatively, one or more process blocks of FIG. **6** may be performed by one or more components of device **500**, such as processor **520**, memory **530**, storage component **540**, input component **550**, output component **560**, and/or communication component **570**.

[0063] As shown in FIG. **6**, process **600** may include inserting a front portion of a transport carrier through a first opening of a docking device and into a chamber of the docking device (block **610**). For example, the load port **104** may insert a front portion of a transport carrier **112** through a first opening **126** of a docking device **108** and into a chamber **132** of the docking device, as described above.

[0064] As further shown in FIG. **6**, process **600** may include contracting a sealing component of the docking device around the transport carrier to form an air-tight seal around the transport carrier (block **620**). For example, the docking device **108** may contract a sealing component **138** of the docking device **108** around the transport carrier **112** to form an air-tight seal around the transport carrier **112**, as described above.

[0065] As further shown in FIG. **6**, process **600** may include, after forming the air-tight seal around the transport carrier removing a carrier door of the transport carrier and opening a chamber door of the docking device to permit access to the transport carrier from the chamber (block **630**). For example, the docking device **108** may, after forming the air-tight seal around the transport carrier **112**, remove a carrier door **140** of the transport carrier **112** and open a chamber door **136** of the docking device **108** to permit access to the transport carrier **112** from the chamber **132**, as described above.

[0066] As further shown in FIG. **6**, process **600** may include accessing, using a wafer transport tool, the transport carrier through the chamber while the air-tight seal is around the transport carrier (block **640**). For example, the interface tool **106** may access, using a wafer transport tool **124**, the transport carrier **112** through the chamber **132** while the air-tight seal is around the transport carrier

112, as described above.

[0067] Process **600** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0068] In a first implementation, accessing the transport carrier **112** includes retrieving (e.g., using the wafer transport tool **124**) a wafer **406** from the transport carrier **112** through the chamber **132** while the air-tight seal is around the transport carrier **112**, and providing (e.g., using the wafer transport tool **124**) the wafer **406** to a semiconductor processing tool **102**. In a second implementation, alone or in combination with the first implementation, accessing (e.g., using the wafer transport tool **124**) the transport carrier **112** includes retrieving a wafer **406** from a semiconductor processing tool **102**, and providing (e.g., using the wafer transport tool **124**) the wafer **406** to the transport carrier **112** through the chamber **132** while the air-tight seal is around the transport carrier **112**.

[0069] In a third implementation, alone or in combination with one or more of the first and second implementations, inserting the front portion of the transport carrier **112** through the first opening **126** includes placing the transport carrier **112** on a shuttle platform **110**, and sliding the shuttle platform **110** with the transport carrier **112** thereon through the first opening **126**. In a fourth implementation, alone or in combination with one or more of the first through third implementations, removing the carrier door **140** includes pressing a mounting plate **302**, attached to the chamber door **136**, against the carrier door **140**, forming a vacuum seal between the mounting plate **302** and the carrier door **140**, and retracting the mounting plate **302** toward the chamber door **136** while the vacuum seal is between the mounting plate **302** and the carrier door **140**.

[0070] In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, the sealing component **138** includes two or more portions (e.g., the portion **138a**, the portion **138b**, the portion **138c**, and/or the portion **138d**), and contracting the sealing component **138** around the transport carrier **112** to form the air-tight seal around the transport carrier **112** includes moving the two or more portions from a first position (**240**) to a second position (**250**), in which the two or more portions are contracted around the transport carrier **112**, to form the air-tight seal around the transport carrier **112**. In a sixth implementation, alone or in combination with one or more of the first through fifth implementations, opening the chamber door **136** to permit access to the transport carrier **112** from the chamber **132** includes sliding the chamber door **136** away from a door frame **134** in the chamber **132**, where sliding the chamber door **136** away from the door frame **134** releases another air-tight seal between the chamber door **136** and the door frame **134**, and sliding the chamber door **136** downward in the chamber **132**.

[0071] Although FIG. **6** shows example blocks of process **600**, in some implementations, process **600** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **6**. Additionally, or alternatively, two or more of the blocks of process **600** may be performed in parallel.

[0072] FIG. **7** is a flowchart of an example process **700** associated with accessing a transport carrier. In some implementations, one or more process blocks of FIG. **7** may be performed by a docking device (e.g., docking device **108**). In some implementations, one or more process blocks of FIG. **7** may be performed by another device or a group of devices separate from or including the docking device, such as a semiconductor processing tool **102**, a load port **104**, an interface tool **106**, a wafer transport tool **124**, and/or another device. Additionally, or alternatively, one or more process blocks of FIG. **7** may be performed by one or more components of device **500**, such as processor **520**, memory **530**, storage component **540**, input component **550**, output component **560**, and/or communication component **570**.

[0073] As shown in FIG. **7**, process **700** may include retrieving, using a wafer transport tool, a wafer from a semiconductor processing tool (block **710**). For example, the wafer transport tool **124** may retrieve the wafer **406** from the semiconductor processing tool **102**, as described above.

[0074] As further shown in FIG. 7, process **700** may include providing, using the wafer transport tool, the wafer to a transport carrier through a chamber of a docking device while an air-tight seal is around a front portion of the transport carrier that is inserted through an opening in the chamber (block **720**). For example, the wafer transport tool **124** may provide the wafer **406** to the transport carrier **114** through the chamber **132** of the docking device **106** while an air-tight seal is around a front portion of the transport carrier **114** that is inserted through the opening **126** in the chamber **132**, as described above.

[0075] As further shown in FIG. 7, process **700** may include placing a carrier door onto the transport carrier while the air-tight seal is around the front portion of the transport carrier (block **730**). For example, the docking device **108** may place the carrier door **140** onto the transport carrier **114** while the air-tight seal is around the front portion of the transport carrier **114**, as described above.

[0076] As further shown in FIG. 7, process **700** may include retracting a sealing component of the docking device around the transport carrier to remove the air-tight seal around the transport carrier after placing the carrier door onto the transport carrier (block **740**). For example, the docking device **108** may retract the sealing component **138** of the docking device **108** around the transport carrier **114** to remove the air-tight seal around the transport carrier **114** after placing the carrier door **140** onto the transport carrier **114**, as described above.

[0077] Process **700** may include additional implementations, such as any single implementation or any combination of implementations described below and/or in connection with one or more other processes described elsewhere herein.

[0078] In a first implementation, placing the carrier door **140** onto the transport carrier **114** includes placing the carrier door **140** onto the transport carrier **114** using the mounting plate **302** on the chamber door **136**. In a second implementation, alone or in combination with the first implementation, the mounting plate **302** includes one or more vacuum holes **306** configured to form a vacuum seal between the carrier door **140** and the mounting plate **302**, and the chamber door **136** further includes a support structure **304** configured to extend the mounting plate **302** toward the transport carrier **114** to place the carrier door **140** onto the transport carrier **114** while the vacuum seal is formed between the carrier door **140** and the mounting plate **302** to place the carrier door **140** onto the transport carrier **114**.

[0079] In a third implementation, alone or in combination with one or more of the first and second implementations, placing the carrier door **140** onto the transport carrier **114** includes latching the carrier door **140** onto the transport carrier **114** using a latch key **308** on the mounting plate **302**. In a fourth implementation, alone or in combination with one or more of the first through third implementations, process **700** includes forming another air-tight seal between the chamber door **136** and a door frame **134** in the chamber **132**. In a fifth implementation, alone or in combination with one or more of the first through fourth implementations, forming the other air-tight seal between the chamber door **136** and the door frame **134** includes forming the other air-tight seal between the chamber door **136** and the door frame **134** prior to retracting the sealing component **138** to remove the air-tight seal around the transport carrier **114**.

[0080] Although FIG. 7 shows example blocks of process **700**, in some implementations, process **700** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. 7. Additionally, or alternatively, two or more of the blocks of process **700** may be performed in parallel.

[0081] In this way, a transport carrier docking device may be positioned between a load port and an interface tool to reduce and/or minimize cross contamination of semiconductor wafers that are transferred between the load port and an associated semiconductor processing tool. The transport carrier docking device may be capable of forming an air-tight seal around a transport carrier while a front portion of the transport carrier is inserted into a chamber of the transport carrier docking device. Semiconductor wafers in the transport carrier may be accessed by a transport tool while the

air-tight seal around the transport carrier prevents and/or reduces the likelihood that contaminants in the semiconductor fabrication facility will reach the semiconductor wafers. The air-tight seal around the transport carrier may reduce defects of the semiconductor wafers that might otherwise be caused by the contaminants, may increase manufacturing yield and quality in the semiconductor fabrication facility, and/or may permit the continued reduction in device and/or feature sizes of integrated circuits and/or semiconductor devices that are to be formed on semiconductor wafers.

[0082] As described in greater detail above, some implementations described herein provide a docking device. The docking device includes a chamber. The docking device includes a first opening in a first side of the chamber. The docking device includes a second opening in a second side of the chamber. The docking device includes a chamber door configured to form a first air-tight seal against a door frame of a docking interface, where the chamber door is configured to form the first air-tight seal in the chamber between the first opening and the second opening when the chamber door is in a closed position. The docking device includes a sealing component configured to form a second air-tight seal around a transport carrier when the transport carrier is at least partially inserted into the first opening.

[0083] As described in greater detail above, some implementations described herein provide a method. The method includes inserting a front portion of a transport carrier through a first opening of a docking device and into a chamber of the docking device. The method includes contracting a sealing component of the docking device around the transport carrier to form an air-tight seal around the transport carrier. The method includes, after forming the air-tight seal around the transport carrier, removing a carrier door of the transport carrier and opening a chamber door of the docking device to permit access to the transport carrier from the chamber. The method includes accessing, using a wafer transport tool, the transport carrier through the chamber while the air-tight seal is around the transport carrier.

[0084] As described in greater detail above, some implementations described herein provide a method. The method includes retrieving, using a wafer transport tool, a wafer from a semiconductor processing tool. The method includes providing, using the wafer transport tool, the wafer to a transport carrier through a chamber of a docking device while an air-tight seal is around a front portion of the transport carrier that is inserted through an opening in the chamber. The method includes placing a carrier door onto the transport carrier while the air-tight seal is around the front portion of the transport carrier. The method includes retracting a sealing component of the docking device around the transport carrier to remove the air-tight seal around the transport carrier after placing the carrier door onto the transport carrier.

[0085] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A docking device, comprising: a chamber; an opening in the chamber; and a sealing component configured to form an air-tight seal around a transport carrier when the transport carrier is at least partially inserted into the opening, wherein a portion of the sealing component is configured to transition downward from a first position to a second position to form the air-tight seal, and wherein the portion is configured to transition horizontally inward from the first position to the second position to form the air-tight seal.

2. The docking device of claim 1, wherein the portion of the sealing component is configured to transition downward and horizontally inward from the first position to the second position in a diagonal path of travel.
3. The docking device of claim 1, wherein the portion of the sealing component is configured to cover an entire top and an entire side of the transport carrier.
4. The docking device of claim 1, wherein another portion of the sealing component is configured to transition upward and horizontally inward from a third position to a fourth position to form the air-tight seal.
5. The docking device of claim 4, wherein the other portion of the sealing component is configured to transition upward and horizontally inward from the third position to the fourth position in a diagonal path of travel.
6. The docking device of claim 4, wherein the other portion of the sealing component is configured to cover an entire bottom and an entire side of the transport carrier.
7. The docking device of claim 1, wherein another portion of the sealing component is configured to transition downward and horizontally inward from a third position to a fourth position to form the air-tight seal.
8. A docking device, comprising: a chamber; an opening in the chamber; and a sealing component configured to form an air-tight seal around a transport carrier when the transport carrier is at least partially inserted into the opening, wherein a first portion of the sealing component is configured to transition downward from a first position to a second position to form the air-tight seal, and wherein a second portion is configured to transition upward from a third position to a fourth position to form the air-tight seal.
9. The docking device of claim 8, wherein the first portion is configured to cover an entire top of the transport carrier.
10. The docking device of claim 9, wherein the first portion is further configured to cover a portion of a first side of the transport carrier.
11. The docking device of claim 10, wherein the first portion is further configured to cover a portion of a second side of the transport carrier opposing the first side.
12. The docking device of claim 8, wherein the second portion is configured to cover an entire bottom of the transport carrier.
13. The docking device of claim 12, wherein the second portion is further configured to cover a portion of a first side of the transport carrier.
14. The docking device of claim 13, wherein the second portion is further configured to cover a portion of a second side of the transport carrier opposing the first side.
15. A docking device, comprising: a chamber; an opening in the chamber; and a sealing component configured to form an air-tight seal around a transport carrier when the transport carrier is at least partially inserted into the opening, wherein a first portion of the sealing component is configured to transition horizontally inward in a first direction from a first position to a second position to form the air-tight seal, and wherein a second portion is configured to transition horizontally inward in a second direction from a third position to a fourth position to form the air-tight seal.
16. The docking device of claim 15, wherein the first portion is configured to cover an entire side of the transport carrier.
17. The docking device of claim 16, wherein the first portion is further configured to cover a portion of a top of the transport carrier.
18. The docking device of claim 17, wherein the first portion is further configured to cover a portion of a bottom of the transport carrier.
19. The docking device of claim 15, wherein the second portion is configured to cover an entire side of the transport carrier.
20. The docking device of claim 19, wherein the second portion is further configured to cover a portion of a top of the transport carrier and a portion of a bottom of the transport carrier.

